

Semi-circular deflector

As per newton's second law of motion, the force of the jet strikes a plate is equal to the rate of change of momentum of the jet, therefore; For the semi-circular deflector ($\alpha = 0^\circ$, $\beta = 120^\circ$ or 180°):

$$\mathbf{F} = \text{Mass of water flowing} \times \text{Change of velocity}$$

So,

$$\mathbf{F} = \rho \mathbf{A} \mathbf{V} \times \Delta \mathbf{V} = \rho \mathbf{A} \mathbf{V} \times (\mathbf{V} \cos(\alpha) - \mathbf{V} \cos(\beta))$$

Therefore:

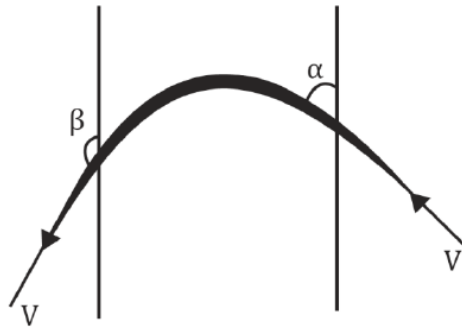
$$\mathbf{F} = \rho \mathbf{A} \mathbf{V}^2 (1 - \cos(\beta))$$

where:

$$\mathbf{V}^2 = \mathbf{V}_0^2 - 2g\mathbf{S}$$

and,

$$\mathbf{V}_0 = \mathbf{Q}/\mathbf{A}$$



Where:

$\mathbf{F}$ = the force exerted by the jet in N.

ρ = density of water in kg/m^3 .

$\mathbf{A}$ = cross sectional area of the jet in m^2 .

$\mathbf{V}$ = actual velocity at the center of the jet in m/s .

$\Delta \mathbf{V}$ = change of the velocity in m/s .

$\mathbf{V}_0$ = velocity of the jet in m/s .

$\mathbf{S}$ = the distant between the nozzle opening and the deflector opening in m.

$\mathbf{g}$ = is the gravitational acceleration in m/s^2 .

$\mathbf{Q}$ = the flow rate in m^3/s .

Answer:

$$A = 0.0000785 \text{ m}^2, S = 0.035 \text{ m}, g = 9.81 \text{ m/s}^2, \rho = 1000 \text{ kg/m}^3$$

$$\diamond Q = \text{-----} \text{ m}^3/\text{s}.$$

$$\diamond V_o = Q / A = \text{-----} = \text{-----} \text{ m/s}.$$

$$\diamond V^2 = V_o^2 - 2gS = \text{-----} = \text{-----} = \text{m/s}.$$

$$\diamond \beta = \text{-----}^\circ.$$

$$\diamond F_{th} = \rho A V^2 (1 - \cos(\beta)) = \text{-----} = \text{-----} \text{ kg.m/s}^2 \text{ (N)}$$

$$\diamond \text{Fac.} = \text{total weight} = \text{total mass} \times g = \text{-----} \text{ kg.m/s}^2 \text{ (N)}$$

$\diamond$ Plot the graph between F_{th} & $F_{ac.}$ vs. Q :

Q (L/min.)									
F (theoretical)									
F (actual)									

